Fiscal Year:	FY 2005	**Task Last Updated:**	FY 09/17/2007
PI Name:	Paik, Ho Jung Ph.D.		
Project Title:	Short-Range Inverse-Square Law Experiment in Space		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Gravitational & relativistic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	hpaik@physics.umd.edu	**Fax:**	FY 301-405-6087
PI Organization Type:	UNIVERSITY	**Phone:**	301-405-6086
Organization Name:	University of Maryland		
PI Address 1:	Department of Physics, University of Maryland		
PI Address 2:			
PI Web Page:			
City:	College Park	**State:**	MD
Zip Code:	20742	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	10/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Paik, Ho Jung		
COI Name (Institution):	Moody, M. Vol (University of Maryland)		
Grant/Contract No.:	NAG3-2874		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity.		
	The test Newton's law of gravitation with a resolution one million times better than the existing limits at 0.1 mm, the superconducting accelerometer technology will be combined with advantages of the low-g environment of space. The experiment will be cooled to < 2 K and the test masses will be magnetically suspended. The soft, low-loss suspension, combined with a low-noise SQUID, leads to extremely low intrinsic noise in the detector. To minimize Newtonian errors, the experiment employs a near null source, a circular disk of large diameter-to-thickness ratio. Two test masses, also disk-shaped, are suspended on the two sides of the source mass at a distance of 0.1 mm and form a superconducting differential accelerometer. A ground test apparatus is under construction and a laboratory test of the inverse-square law is planned. The objective of ISLES (Inverse-Square Law Experiment in Space) is to perform a null test of Newton's law on ISS with a resolution of 10 ppm at distances ranging from 100 micrometer to 1 mm. This		

Task Description:	represents an improvement by 10^6 over the existing limits at 100 micrometers, and even greater improvement at shorter distances. ISLES will be sensitive enough to detect axions with the strongest allowed coupling and test the string-theory prediction with R > 5 micrometer. To reach this sensitivity, the existing superconducting accelerometer technology will be combined with advantages of the low-g environment of space. The Standard Model of particle physics, although highly successful, has one serious blemish: the so-called "strong CP problem." A very plausible explanation for this phenomenon was proposed over two decades ago, but experimentalists have so far failed to verify it. One ramification of the theory is the prediction of the existence of a new particle, called the "axion," which mediates a short-range mass-mass interaction, as well as a force between intrinsic spin and mass. The axion is also a strong candidate for cold dark matter. Confirmation (or rejection) of this prediction would therefore have a major impact on our understanding of the universe, from its most microscopic constituents to its grand structure. Another impetus for testing the law at the shortest ranges possible comes from string theories. In these theories, the extra spacetime dimensions are compactified, leaving the observed 3+1 spacetime dimensions. For two large extra dimensions, the law of gravity changes from 1/r^2 to 1/r^4, as the separation r is reduced to below the radius of compactification R. An intriguing possibility that R > 1 micrometer has been suggested recently that would imply a violation of the inverse-square law below 1 micrometer distance. This is the first prediction of a string theory that can be tested, and a discovery of this deviation would be ground breaking.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	A. Analysis of metrology errors A detailed analysis has been carried out for source metrology errors to determine the required machining tolerances. The azimuthal density and dimensional errors can be averaged out by rotating the test masses. The radial density variation was expanded as a Fourier Bessel series with random coefficients. The thickness variation was represented as random surface density fluctuations. In spite the proximity of the test mass to the source mass (~ 0.1 mm), high-order terms of up to order 100, which corresponds to a resolution of ~ 4 mm, contribute significantly to the total error. The thickness fluctuations are much more important than the density error. B. Analysis and design of the superconducting circuits The driving and sensing circuits were fully analyzed to optimize their performance and define the inductances needed for the pancake coils and superconducting transformers. Detailed examination and modeling of the source driving resulted in a change in the circuitry. The new circuit uses much smaller inductors and permits cancellation of the nonlinear term, which is important to avoid generating an undesirable Newtonian force at the second harmonic. To reduce the recoil of the apparatus caused by the source motion and the resulting common-mode acceleration, a force will be applied from the cryostat to the suspended housing. For this purpose, two shaker coils have been added on the niobium (Nb) covers, as well as a shaker driving circuit. C. Construction of the apparatus Two 1.6-mm thick tantalum (Ta) sheets were "two face grounded" commercially to improve the surface flatness. A measurement with dial indicators showed the surface to be flat to 10 microns, which is an acceptable tolerance for the ground experiment. A source mass was produced from one of these sheets by wire EDM and then heat-treated to relieve stress. Four test masses were constructed, also with wire EDM, from a 0.25-mm thick Ta sheet. Two Nb housings that hold the source mass, test masses, and various coils together were machined. A new Nb hot zone had to be constructed for our high-vacuum oven to accommodate the 22.9-cm diameter housings and source mass. Superconducting shields for the test masses were fabricated by diffusion-bonding 12.5-micron Nb sheets to Nb rings. The source driving coils, various sensing coils, and alignment coils were wound. The superconducting transformers were wound and tested. To reduce the susceptibility of the SQUID amplifiers to rf interference, shielded cables for the current charging leads were constructed and added to the cryostat. The 1/r^2 law apparatus needs to be cooled to 2 K to reduce its sensitivity to temperature fluctuation. The cryostat was cooled down several times with the impedance of the capillary for the 2-K stage varied until a stable operation was achieved near 2 K. The entire apparatus was assembled and integrated with the cryostat. This required careful mechanical alignment and assembly of the source and test masses as well as coils and shields, complicated routing of dozens of leads for various circuits, and making scores of spot-welded superconducting joints. The entire apparatus is undergoing the cryogenic test.
Bibliography Type:	Description: (Last Updated: 10/30/2007)
Abstracts for Journals and Proceedings	Paik, H. J. "Probing extra dimensions using a superconducting accelerometer" Joint Korean Physical Society/Association of Korean Physicists in America Meeting, Seoul, Korea Bulletin of KPS Meeting, April, 2005 , Apr-2005
Articles in Peer-reviewed Journals	Paik, H. J.; Moody, M. V.; Strayer, D. M. "Short-range inverse square law experiment in space" Gen. Rel. Grav. 36, 523-537 (2004). , Mar-2004
Articles in Peer-reviewed Journals	Paik,H. J.; Prieto, V.: Moody, M. V. "Probing extra dimensions using a superconducting accelerometer" J. Kor. Phys. Soc. 45, S104-S109 (2004) , Oct-2004
Awards	Paik, H. J. "American Physical Society Fellowship" Nov-2004
Presentation	Paik, H. J. "Probing Extra Dimensions Using a Superconducting Gravity Gradiometer" CaJAGWR seminar at California Institute of Technology, Pasadena, California Apr-2004
Presentation	Paik, H. J. "Does Gravity Experience Extra Dimensions" Colloquium at California State University, Long Beach, California Sep-2004

Presentation	Paik, H. J. "Probing Extra Dimensions Using a Superconducting Accelerometer" HEPL seminar at Stanford University, Stanford, California Nov-2004
Presentation	Paik, H. J. "Does Gravity Experience Extra Dimensions?" Colloquium at National Astronomical Observatory, Tokyo, Japan Apr-2005